

CSC1060 - Data Analytics

Fundamentals Problem Solving 2 -

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Sample Document for student reference

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1. Introduction

Student performance is an important area of interest for educational institutes because it helps identify factors that affect academic success and supports informed decision making. Predictive analytics uses past data and machine learning techniques to identify patterns and predicts future outcomes that assists the educators to identify students who may require academic support.

2. Problem Definition

The aim of this project is to develop and evaluate machine learning regression models that can predict students' final grades using demographic, attendance, student behaviour and previous academic performance data. The project follows the data analytics lifecycle which includes data preparations, exploratory data analysis (EDA), feature engineering, model development and model evaluation.

The data used in this project contains information on student characteristics such as school, student sex, age, home address type, weekly study time, past class failures, school absences, first period grade, second period grade and final grades.

3. Data Preparation

The student performance dataset was loaded into Jupyter Notebook using the Pandas library for analysis. Originally, the data contained 649 records and 10 variables that featured demographic information, student habits, attendance records, previous academics performance and the final grade. Early assessment was carried out using functions such as 'head()', 'info()' and 'describe()' to understand the structure of the date, its types and summary statistics of the dataset.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 649 entries, 0 to 648
Data columns (total 10 columns):
#   Column              Non-Null Count  Dtype
---  -
0   School              649 non-null    object
1   Student Sex         649 non-null    object
2   Age                 649 non-null    int64
3   Home Address Type   649 non-null    object
4   Weekly Study Time   649 non-null    int64
5   Past Class Failures 649 non-null    int64
6   School Absences     649 non-null    int64
7   First Period Grade  649 non-null    int64
8   Second Period Grade 649 non-null    int64
9   Final Grade         649 non-null    int64
dtypes: int64(7), object(3)
memory usage: 50.8+ KB
```

The dataset was checked for data quality issues and a review of missing values showed that there were no missing values in any of the variables which meant that the dataset was complete and did not require removal. Duplicate records were also found. Eight duplicate rows were found and removed to prevent repeated observations from affecting the machine learning models. After removing the duplicates, the dataset contained 641 unique records.

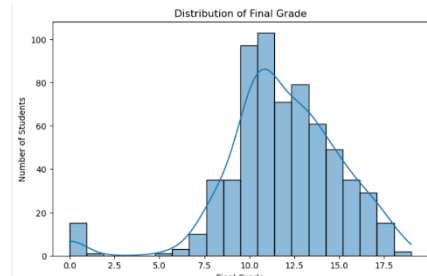
	Age	Weekly Study Time	Past Class Failures	School Absences	First Period Grade	Second Period Grade	Final Grade
count	649.000000	649.000000	649.000000	649.000000	649.000000	649.000000	649.000000
mean	16.744222	1.930663	0.221880	3.659476	11.399076	11.570108	11.906009
std	1.218138	0.829510	0.593235	4.640759	2.745265	2.913639	3.230656
min	15.000000	1.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	16.000000	1.000000	0.000000	0.000000	10.000000	10.000000	10.000000
50%	17.000000	2.000000	0.000000	2.000000	11.000000	11.000000	12.000000
75%	18.000000	2.000000	0.000000	6.000000	13.000000	13.000000	14.000000
max	22.000000	4.000000	3.000000	32.000000	19.000000	19.000000	19.000000

	School	Student Sex	Age	Home Address Type	Weekly Study Time	Past Class Failures	School Absences	First Period Grade	Second Period Grade	Final Grade
0	Gabriel Pereira	Female	18	Urban	2	0	4	0	11	11
1	Gabriel Pereira	Female	17	Urban	2	0	2	9	11	11
2	Gabriel Pereira	Female	15	Urban	2	0	6	12	13	12
3	Gabriel Pereira	Female	15	Urban	3	0	0	14	14	14
4	Gabriel Pereira	Female	16	Urban	2	0	0	11	13	13
5	Gabriel Pereira	Male	16	Urban	2	0	6	12	12	13
6	Gabriel Pereira	Male	16	Urban	2	0	0	13	12	13
7	Gabriel Pereira	Female	17	Urban	2	0	2	10	13	13
8	Gabriel Pereira	Male	15	Urban	2	0	0	15	16	17
9	Gabriel Pereira	Male	15	Urban	2	0	0	12	12	13

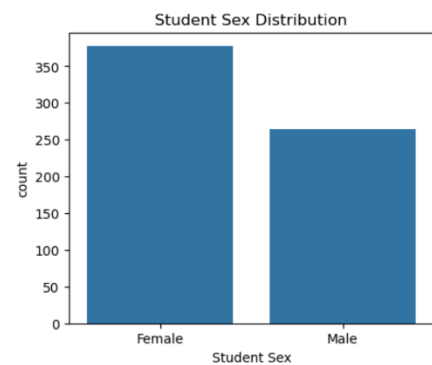
4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the dataset and identify the relationships between variables before developing the machine learning models. The concepts were created using Matplotlib and Seaborn to examine data distributions and correlations.

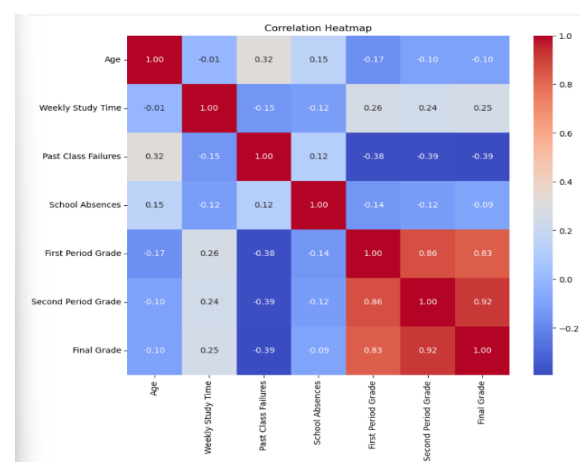
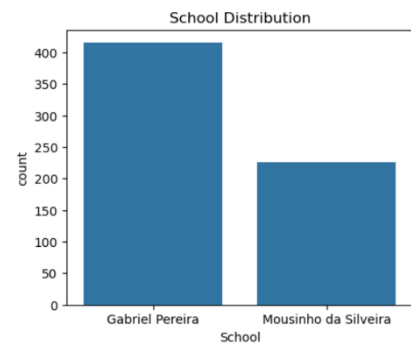
The distribution of the Final Grade variable showed that student grades followed an approximately normal distribution, with most students achieving grades between 9 and 14. Only a small number of students achieved very low or very high grades which indicates that the dataset was relatively balanced for regression modelling.



The distribution of students by Student Sex showed that the dataset contained more female students than male students. The school distribution shows that a larger number of students attended Gabriel Pereira School rather than Mousinho da Silveira School. While these categorical variables give us useful descriptive information, they were not expected to have as much influence on final grades as previous academic performance.



A correlation heatmap was produced and the analysis showed that Second Period Grade had the strongest positive correlation with Final Grade, followed by First Period Grade. This implies that previous academic performance is the greatest indicator of a students' result. Past Class Failures showed a moderate negative correlation with Final Grade, meaning that the students with more previous failures generally achieved lower final grades. Weekly Study Time had a weak positive relationship with Final Grade, whereas Age and School Absence only showed weak correlations.



5. Experiment

5.1 Processing

Feature engineering was performed to prepare the dataset for machine learning regression models. The dataset contained three categorical variables which was School, Student Sex and Home Address Type. Because machine learning algorithms require numerical input, the categorical variables were converted into numerical values using one-hot encoding. The 'get_dummies()' function in Pandas was used with the 'drop_first=True' by removing one category from each encoded variable.

	Age	Weekly Study Time	Past Class Failures	School Absences	First Period Grade	Second Period Grade	Final Grade	School_Mousinho da Silveira	Student Sex_Male	Home Address Type_Urban
0	18	2	0	4	0	11	11	False	False	True
1	17	2	0	2	9	11	11	False	False	True
2	15	2	0	6	12	13	12	False	False	True
3	15	3	0	0	14	14	14	False	False	True
4	16	2	0	0	11	13	13	False	False	True

After encoding the categorical variables, the dataset was divided into two components. The feature matrix (x) and the target variable (y). The feature matrix included predictor variables while Final Grade was chosen as the target variable to be predicted by the regression models. The dataset was then split into training (80%) and testing (20%) subsets using the 'train_test_split()' function. To end, feature scaling was used using the 'StandScaler' algorithm. Scaling standardises numerical variables by transforming them to have a mean of 0 and standard deviation of 1.

(641, 9)

(641,)

Training set: (512, 9)

Testing set: (129, 9)

5.2 Models

Three supervised learning regression models were built and compared to predict students' final grades. Model performance was calculated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the Coefficient of Determination (R^2 Score).

The first model applied was Linear Regression. This assumes a linear relationship between the predictor variables and the target variable. It was selected because it is a simple and widely used model for regression problems. It presents an effective benchmark against which complex machine learning models can be compared.

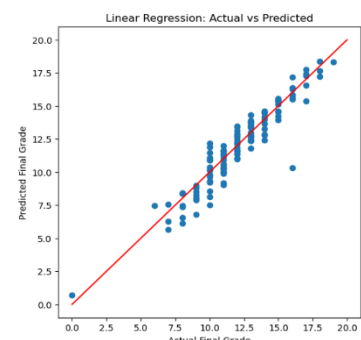
Linear Regression Results

MAE : 0.76

MSE : 1.02

RMSE : 1.01

R^2 : 0.88



The second model was Decision Tree Regression. It can capture non-linear relationships by repeatedly splitting the data into smaller groups based on feature values. This allows the model to identify more complex patterns within the student performance data.

Decision Tree Results

MAE : 0.94
MSE : 2.44
RMSE : 1.56
R² : 0.71

The third model was Random Forest Regression which is an accumulate learning model that combines predictions from multiple decision tress to improve predictive accuracy and reduce the risk of overfitting and by averaging the predictions of many trees, the models usually produce more reliable results than a single decision tree.

Random Forest Results

MAE : 0.82
MSE : 1.47
RMSE : 1.21
R² : 0.82

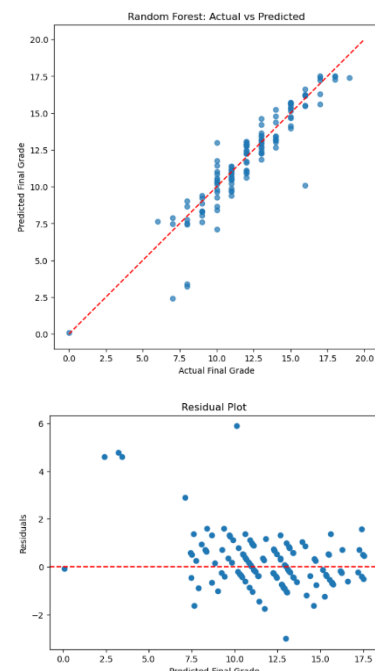
5.3 Evaluation

The performance of the three regression models was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the Coefficient of Determination (R² Score). These were selected because they provide a detailed assessment of prediction accuracy. MAE measures the average difference between predicted and actual values, RMSE gives greater weight to larger prediction errors and the R² score indicates how well the model explains the variation in the target variable.

	Model	MAE	RMSE	R2 Score
0	Linear Regression	0.760667	1.010790	0.877739
1	Decision Tree	0.941860	1.560784	0.708492
2	Random Forest	0.820891	1.214417	0.823518

The Best Performing Model achieved the highest prediction accuracy as indicated by its evaluation metrics. It produced the highest R² and the lowest prediction errors showing a stronger ability to predict students' final grades than the other models.

To further evaluate model performance, an actual vs predicted plot was created. The results showed that most predicted values were close to that actual students' grade which means that the selected model was able to produce reliable predictions. A residual plot was also tested and showed that prediction errors were commonly distributed around zero.



6. Findings

The results of this analysis project convey that previous academic performance is the strongest indicator of a student's final grade. The exploratory data analysis showed that First Period Grade and Second Period Grade has the strongest positive relationships with the final grade, while Past Class Failures showed a moderate negative relationship. These results indicate that students who perform well in earlier assessment periods are highly likely to achieve higher final grades while students with history of failing previous classes are at bigger risk of lower academic performance.

The machine learning models has confirmed that student performance can be predicted with a high level of accuracy using demographic, attendance, study behaviour and previous academic achievement data. Comparing multiple regression models made it possible to recognise the most suitable algorithm for predicting final grades.

From an educational perspective, predictive analytics can support early intervention strategies. Students identified as being at risk of poor academic performance can receive additional learning support such as tutoring or academic counselling before final assessments. Early recognition allows educators to make informed decisions that could help improve student success.